Model Evaluation Report

Crypto Volatility Spike Detection — BTC-USD

1. Dataset Summary

Property	Value
Source	Coinbase Advanced Trade WebSocket (BTC-USD)
Collection Period	April 5, 2026, ~20 minutes
Total Feature Rows	5,261 (labeled)
Split Method	Time-based: 60/20/20
Train / Val / Test	3,156 / 1,052 / 1,053
Spike Rate (Train)	15.7%
Spike Rate (Val)	0.0%
Spike Rate (Test)	2.8%
Threshold	0.00005787 (90th percentile)

2. Model Comparison

Model	PR-AUC	F1@threshold
Baseline (Z-Score)	0.0164	0.0000
XGBoost	0.0438	0.0839

3. Analysis

XGBoost outperforms the baseline by 2.7x on PR-AUC (0.0438 vs 0.0164). Both models show low absolute performance due to limited training data (~20 minutes) and significant temporal distribution shift — the spike rate drops from 15.7% in training to 0.0% in validation and 2.8% in test.

The XGBoost model relies heavily on tick intensity features (trade frequency), which captured 96.8% of total feature importance. This suggests that trading activity frequency is a stronger short-term volatility predictor than price-based features in this dataset.

4. Inference Performance

The model scores 1,053 test rows in 0.0034 seconds — 61,434x faster than real-time. This far exceeds the 2x real-time requirement, confirming the model is suitable for online deployment.

5. Drift Analysis

Kolmogorov-Smirnov tests between early and late data windows detected drift in 21/21 features ($p < 0.05$). This is expected given the short collection period and natural market microstructure changes. In production, this would trigger model retraining or threshold recalibration.

6. Recommendations

1. Collect data over multiple days to capture diverse volatility regimes.
2. Add more trading pairs (ETH-USD, SOL-USD) for generalization.
3. Experiment with sequence models (LSTM) to capture temporal dependencies.
4. Implement online learning to adapt to market regime changes.
5. Add order flow and on-chain features for richer signal.